Generation and Analysis of Frequency Modulated (FM) and of Phase Modulated (PM) Signals using MATLAB

Aim: To visualize the electric field distribution of the dominant TE_{10} mode in a rectangular waveguide using MATLAB.

```
clc;
clear;
close all;

%% Waveguide dimensions (meters)
% Example dimensions; change these to match your waveguide.
a = 22.86e-3;   % Width
b = 10.16e-3;   % Height

%% Field and grid settings
E0 = 1;         % Peak electric field

Nx = 200;       % Number of points along x
Ny = 100;       % Number of points along y

x = linspace(0, a, Nx);
y = linspace(0, b, Ny);

% X and Y are matrices for evaluating the field over the cross-section.
[X, Y] = meshgrid(x, y);

%% TE10 electric field
% TE10 varies across the waveguide width (x) and is constant along y.
```

```
Ey_TE10 = E0 * sin(pi * X / a);

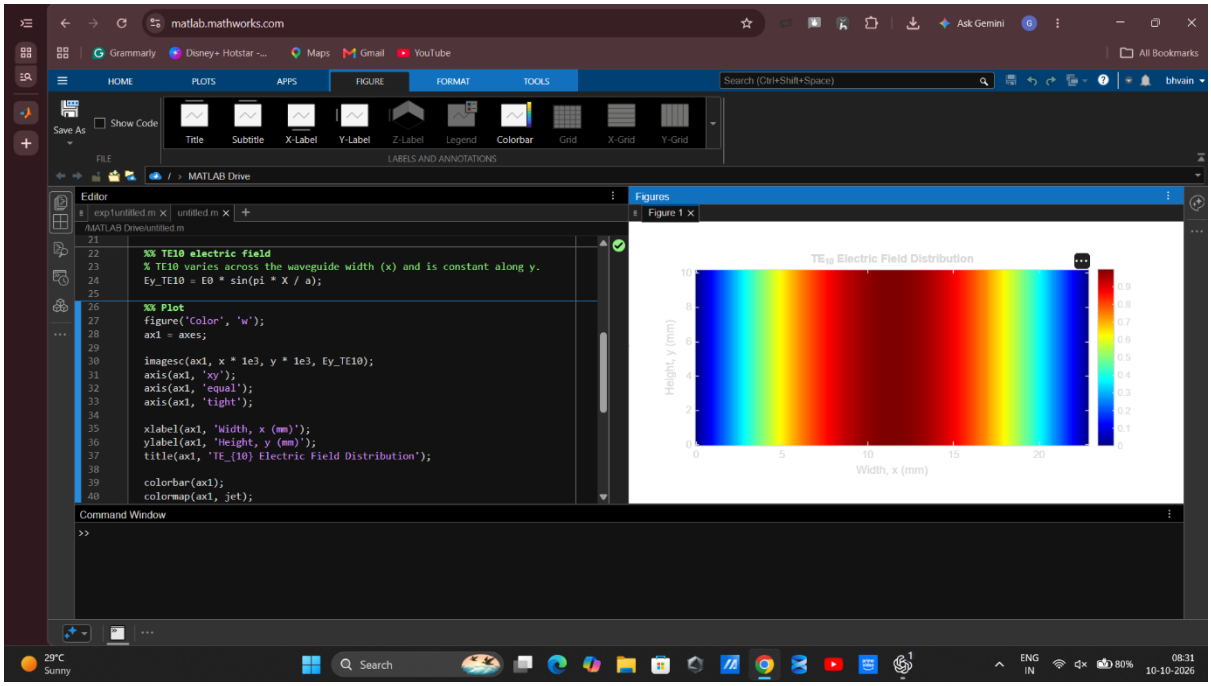

%% Plot
figure('Color', 'w');
ax1 = axes;


imagesc(ax1, x * 1e3, y * 1e3, Ey_TE10);
axis(ax1, 'xy');
axis(ax1, 'equal');
axis(ax1, 'tight');


xlabel(ax1, 'Width, x (mm)');
ylabel(ax1, 'Height, y (mm)');
title(ax1, 'TE_{10} Electric Field Distribution');


colorbar(ax1);
colormap(ax1, jet);
```

OUTPUT :


matlab.mathworks.com
HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
MATLAB Drive
Editor
untitled.m
%% TE10 electric field
% TE10 varies across the waveguide width (x) and is constant along y.
Ey_TE10 = E0 * sin(pi * X / a);
%% Plot
figure('Color', 'w');
ax1 = axes;
imagesc(ax1, x * 1e3, y * 1e3, Ey_TE10);
axis(ax1, 'xy');
axis(ax1, 'equal');
axis(ax1, 'tight');
xlabel(ax1, 'Width, x (mm)');
ylabel(ax1, 'Height, y (mm)');
title(ax1, 'TE_{10} Electric Field Distribution');
colorbar(ax1);
colormap(ax1, jet);
Figures
Figure 1
TE10 Electric Field Distribution
Height, y (mm)
Width, x (mm)
Command Window
>>
29°C
Sunny
Search
ENG
IN
80%
08:31
10-10-2026